

Power Move: Solar Adoption Among Firms

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Abstract

Fossil fuel consumption for electricity generation remains a major contributor to global emissions, yet firm adoption of renewable energy has been slower than necessary to achieve climate goals. This paper investigates what drives firm-level adoption of distributed solar generation by leveraging unique microdata on solar panel installations across firms. To identify the causal effects of economic incentives on adoption decisions, I exploit variation generated by a net-metering scheme that differentially affected firms' cost-benefit calculations for solar investment, finding evidence of a positive substitution effect. To quantify the relative importance of different price and cost channels, I estimate a discrete-continuous choice model that jointly captures both the adoption decision and the capacity choice. I find that while energy price subsidies did account for a significant increase in solar adoption, firms appear to be more sensitive to high upfront investment costs. Under an equivalent-cost comparison, an installation cost subsidy would add twice as many adopters and more than double additional installed capacity compared to the baseline energy price subsidy. I calculate that installation subsidies achieve carbon abatement at US\$12 per ton CO₂ — twelve times more cost-effectively than energy price subsidies and well below prevailing carbon prices in international markets — demonstrating that policies targeting upfront investment barriers can deliver emissions reductions at substantially lower cost per ton abated.

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